

Q.32 A circular shaft of 60mm diameter is required to transmit torque is another shaft. Find the safe torque, which the shaft can transmit, if the shear stress cannot exceed 50Mpa.

Q.33 Explain the following terms:

- a) Column
- b) Effective length of Column

Q.34 Write the mathematical expression for calculating the deflection in leaf springs.

Q.35 Explain Moment of Inertia and Second Moment of Area.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

Q.36 Find the Centroid of an I-section whose dimensions are as under:

Top Flange = 10 cm x 4 cm

Web Flange = 4 cm x 10 cm

Bottom Flange = 15 cm x 4 cm

Q.37 A cantilever beam of length 6m carries a uniformly distributed load of 3kN/m which runs over a length of 3m from the free end. In addition to this there is a point load of 1kN at a distance of 3m from the free end. Draw SFD and BMD.

Q.38 What is Screw Jack. Explain its principle of operation and derive the expression for its Mechanical Advantage and Velocity Ratio.

No. of Printed Pages : 4
Roll No.

202021

2nd Year /Advance Diploma in Tool and Die Making Subject:- Applied Mechanics & Strength of Materials

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 Unit of Shear Stress-

- a) N-m
- b) N-m
- c) N/m^2
- d) N/m^3

Q.2 Which of the following is a scalar quantity?

- a) Force
- b) Stress
- c) Volume
- d) Velocity

Q.3 Which of the following is a Fundamental Quantity?

- a) Length
- b) Acceleration
- c) Weight
- d) Momentum

Q.4 The forces which meet at one point and their lines of action also lies on the same plane, are known as

- a) Coplanar concurrent forces
- b) Coplanar non-concurrent forces
- c) Non-Coplanar non-concurrent forces
- d) Non-Coplanar concurrent forces

Q.5 The rotational tendency of a force is called

- a) Shear force
- b) Moment
- c) Centroid
- d) Couple

(20)

(4)

202021

(1)

202021

- Q.6 The coefficient of friction depends upon
 a) Area of contact b) Shape of surfaces
 c) Strength of surfaces d) Nature of surfaces
- Q.7 Rate of change of momentum is proportional to the
 a) Displacement b) Velocity
 c) Acceleration d) Applied force
- Q.8 Mathematically, Mechanical Advantage is expressed as
 a) W/P b) P/W
 c) $W+P$ d) $W-P$
- Q.9 The Column that fails due to direct stress is called _____.
 a) Short Column b) Long Column
 c) Medium Column d) Slender Column
- Q.10 When a shaft is subjected to a twisting moment, every cross-section of the shaft will be under _____.
 a) Tensile Stress b) Compressive Stress
 c) Shear Stress d) All of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Full form of SFD is _____.
- Q.12 Define Torsion.
- Q.13 Define Close Coil Helical Spring.
- Q.14 Strength of column depend upon its _____.
- Q.15 Output of a real machine is always _____ than input.

- Q.16 Write the relation between Modulus of Elasticity (E) and Modulus of Rigidity (G).
- Q.17 The S.I. unit of Second moment of Area is _____.
- Q.18 The positive bending moment is known as _____.
- Q.19 Write Bending equation.
- Q.20 Define Proof Resilience.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 State principle of momentum. Also write its mathematical expression.
- Q.22 State the laws of Dynamic Friction.
- Q.23 Explain the condition of equilibrium of a body under non-current forces.
- Q.24 Mention the differences between Kinematics and Kinetics.
- Q.25 Write a short note on:
 (a) Angle of Repose (b) Rolling friction
- Q.26 A car of mass 50 kg gets the speed of 10m/s in 5 seconds from 0 velocity. Calculate the force required by the engine to drive the car.
- Q.27 State the Theorem of Perpendicular Axis.
- Q.28 Explain the uses of simple machines.
- Q.29 Define -
 (a) Resilience (b) Modulus of Resilience
- Q.30 Mention various types of strains.
- Q.31 State four assumption in theory of pure bending.